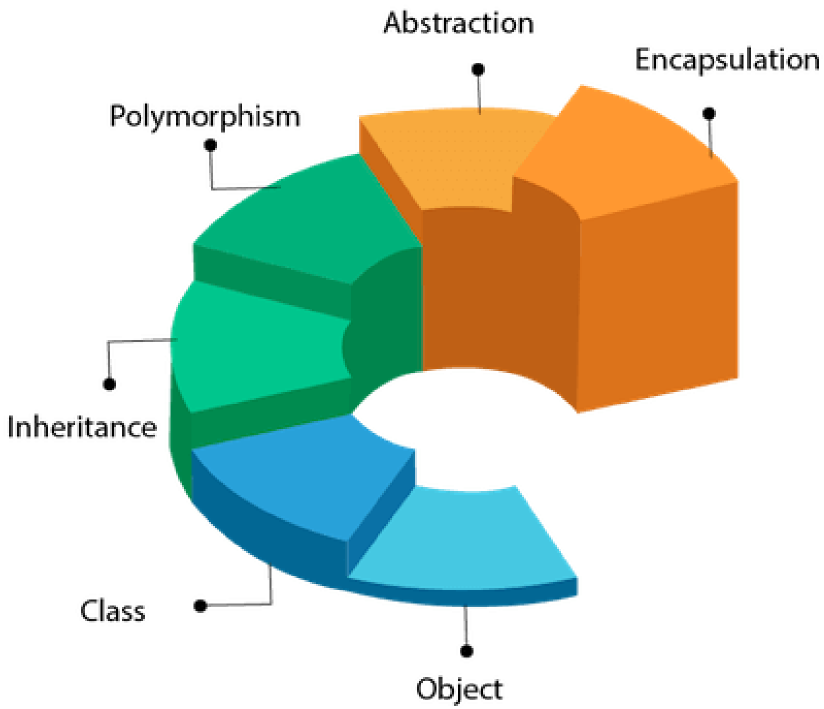
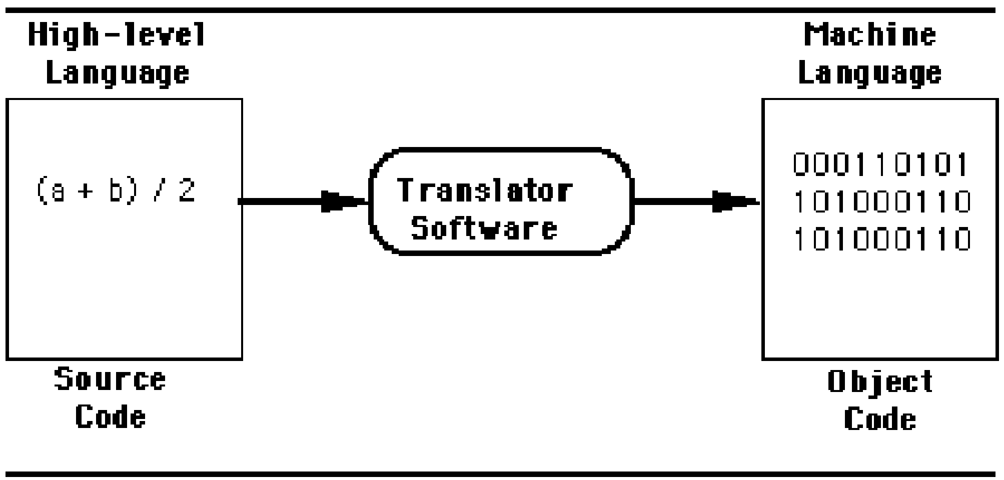


Basic Concepts Of Objects Oriented Programming
Week-1

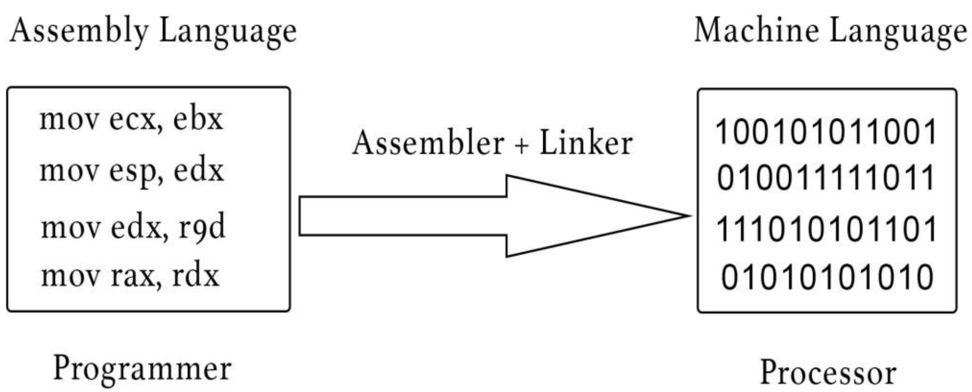
Machine level Language

- Machine code or machine language is a set of instructions executed directly by a computer's central processing unit (CPU).
- instruction performs a very specific task, such as a load, a jump, or an ALU operation on a unit of data in a CPU register or memory.
- Every program directly executed by a CPU is made up of a series of such instructions.



Assembly level Language

- An assembly language (or assembler language) is a low-level programming language for a computer, or other programmable device, in which there is a very strong (generally one-to-one) correspondence between the language and the architecture's machine code instructions.
- Assembly language is converted into executable machine code by a utility program referred to as an assembler; the conversion process is referred to as assembly, or assembling the code.

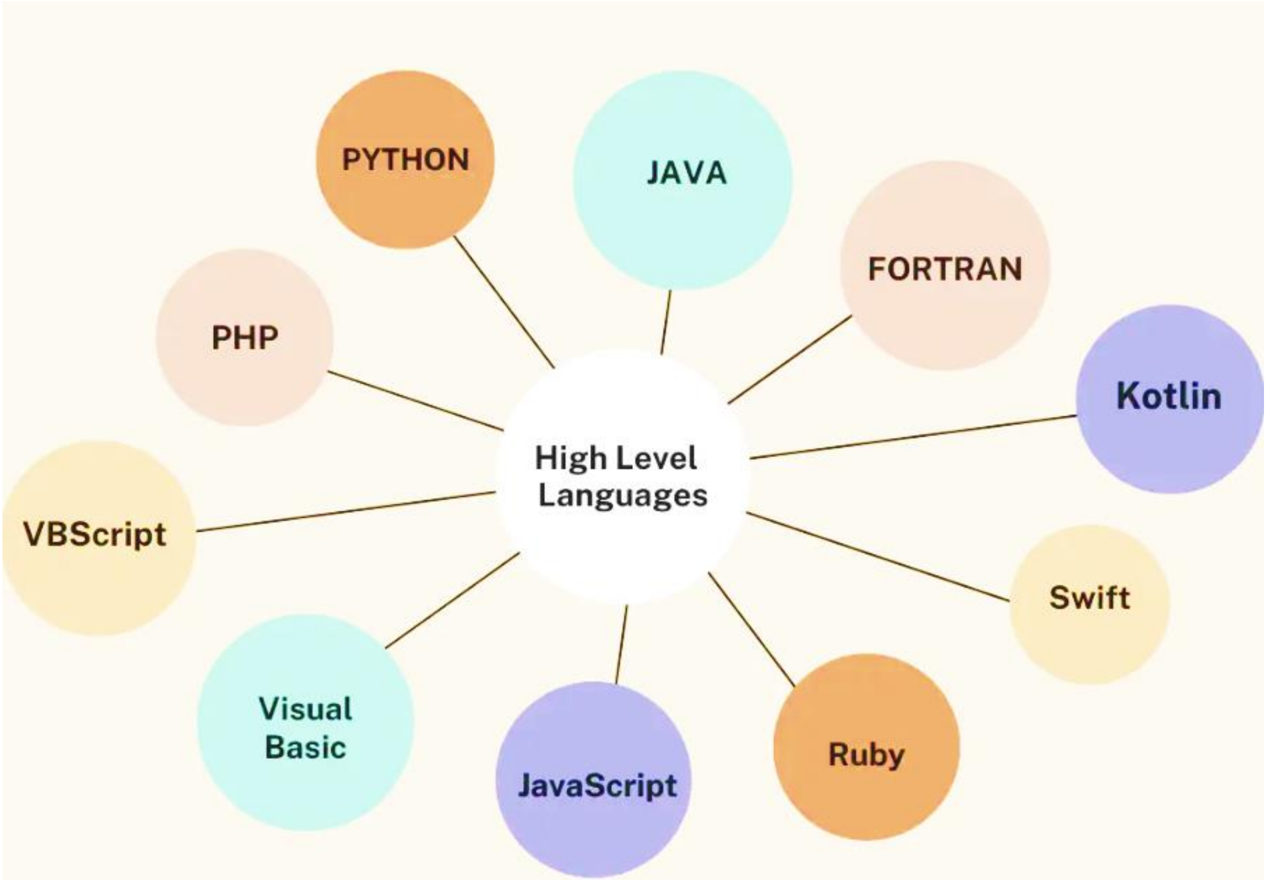


High level Language

- High-level language is any programming language that enables development of a program in much simpler programming context and is generally independent of the computer's hardware architecture.

High level Language

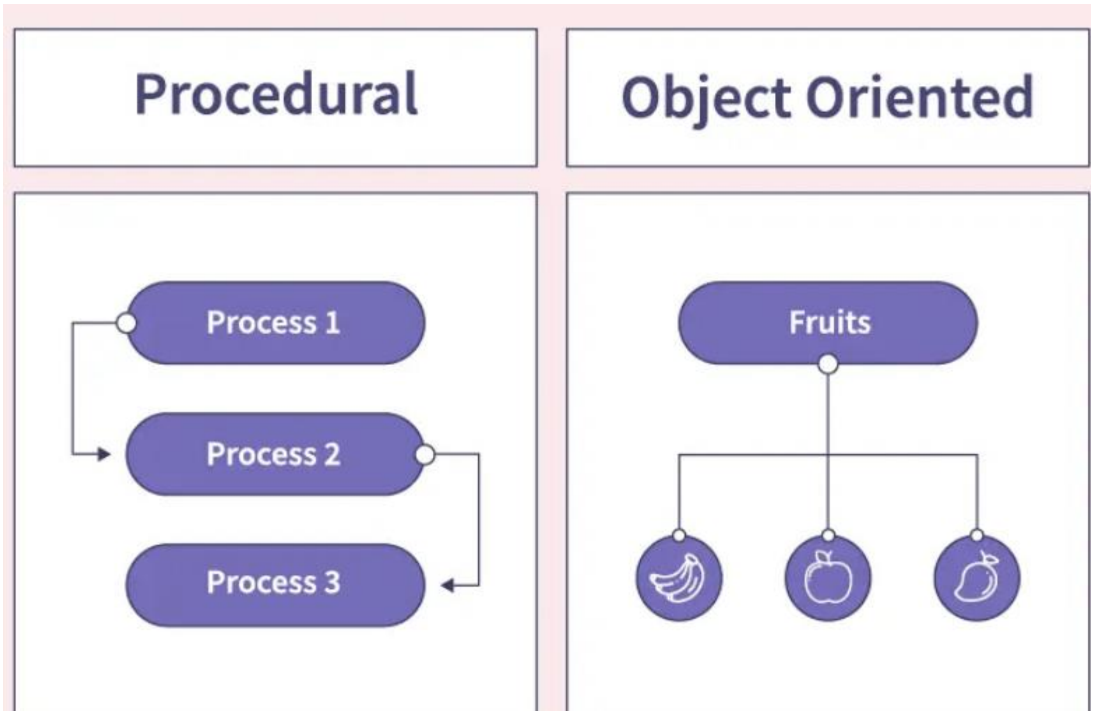
- High-level language has a higher level of abstraction from the computer, and focuses more on the programming logic rather than the underlying hardware components such as memory addressing and register utilization.
- The first high-level programming languages were designed in the 1950s. Now there are dozens of different languages, including Ada , Algol, BASIC, COBOL, C, C++, JAVA, FORTRAN, LISP, Pascal, and Prolog.



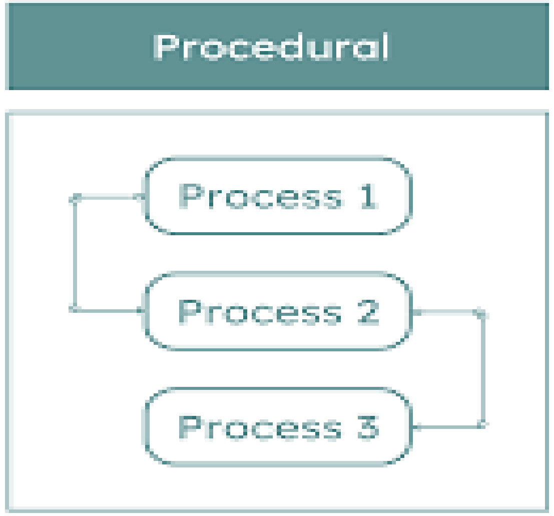
High level Language

The high-level programming languages are broadly categorized in to two categories:

- Procedure oriented programming(POP) language.
- Object oriented programming(OOP) language.

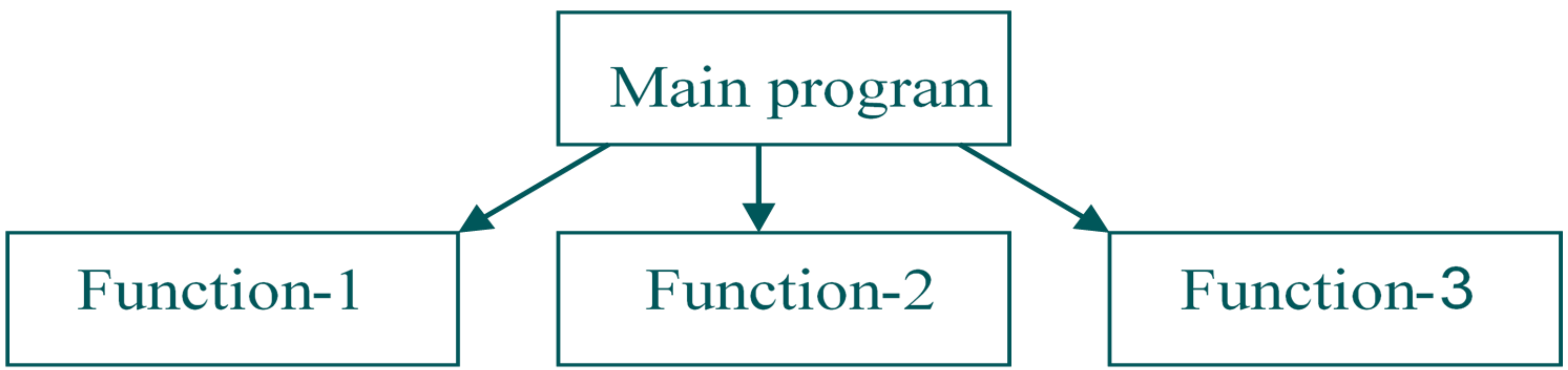


Procedure oriented programming(POP) language



Procedure Oriented Programming Language

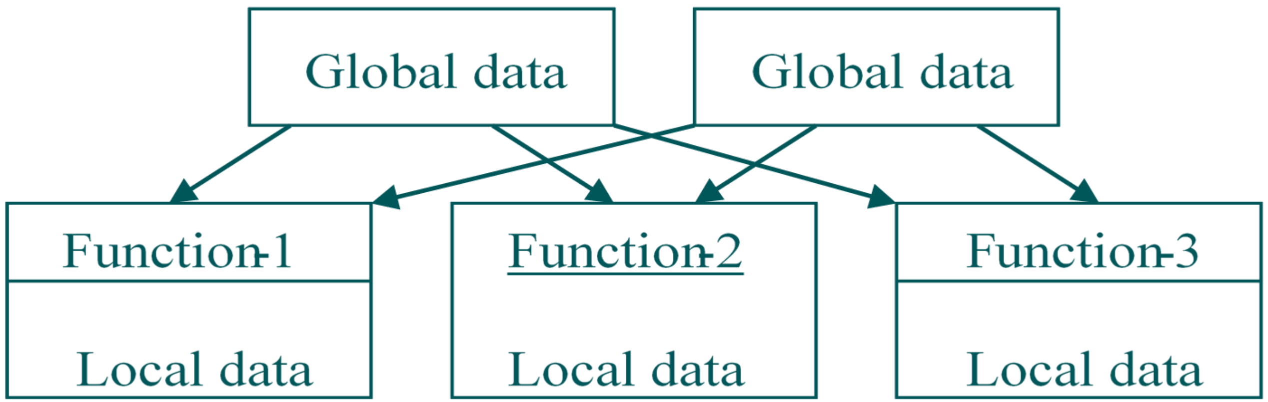
- In the procedure oriented approach, the problem is viewed as sequence of things to be done such as reading , calculation and printing.
- Procedure oriented programming basically consist of writing a list of instruction or actions for the computer to follow and organizing these instruction into groups known as functions.



Procedure Oriented Programming Language

The disadvantage of the procedure oriented programming languages is:

- Global data access
- It does not model real word problem very well
- No data hiding



Structured / Procedure Programming Methodology (SPM)

Structured programming (sometimes known as **modular programming**) is a subset of procedural programming that enforces a logical structure on the written program to make it more efficient, and easier to understand and modify.

Basic features of SPM

- Emphasis on doing **algorithms**.
- Large Programs are divided into smaller programs known as **Functions**
- Most of the function shares **global data**.
- Data move around the system **globally** from function to function.
- Function **transfers** the data from one form to another.
- Employs **top-down** approach of Programming.
- **Example: C, Pascal, FORTRAN**

Problems with Structured Programming Methodology (SPM)

- Reach their limit when project becomes too **large**.
- Large program became more **complex**.
- Functions have **unrestricted** access to **global data**.

Characteristics Of Procedure Oriented Programming

- Emphasis is on doing things(algorithm)
- Large programs are divided into smaller programs known as functions.
- Most of the functions share global data
- Data move openly around the system from function to function
- Function transforms data from one form to another.
- Employs top-down approach in program design